# 1F1B EP All-to-All Overlap

## Hide MoE token exchange behind the neighboring micro-batch

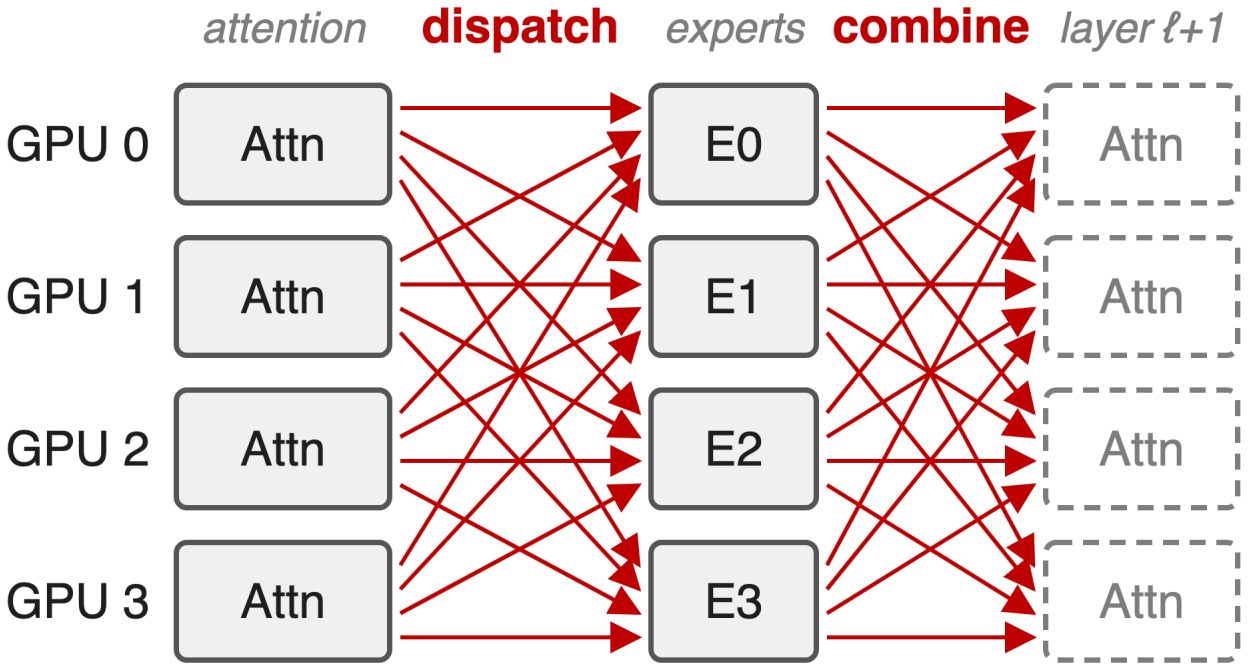
NVIDIA · Megatron-Core MoE (v0.14+)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Comm/Compute Overlap · arxiv.org/abs/2603.07685

# Fine-grained MoE adds two all-to-alls per layer

- DeepSeek-V3: **685B** params, **37B** active
- **256** experts, top-8 routing
- **116** dispatch/combine per forward pass; EP64 spans **8** H100 nodes




*repeats in all 58 MoE layers and again in backward*

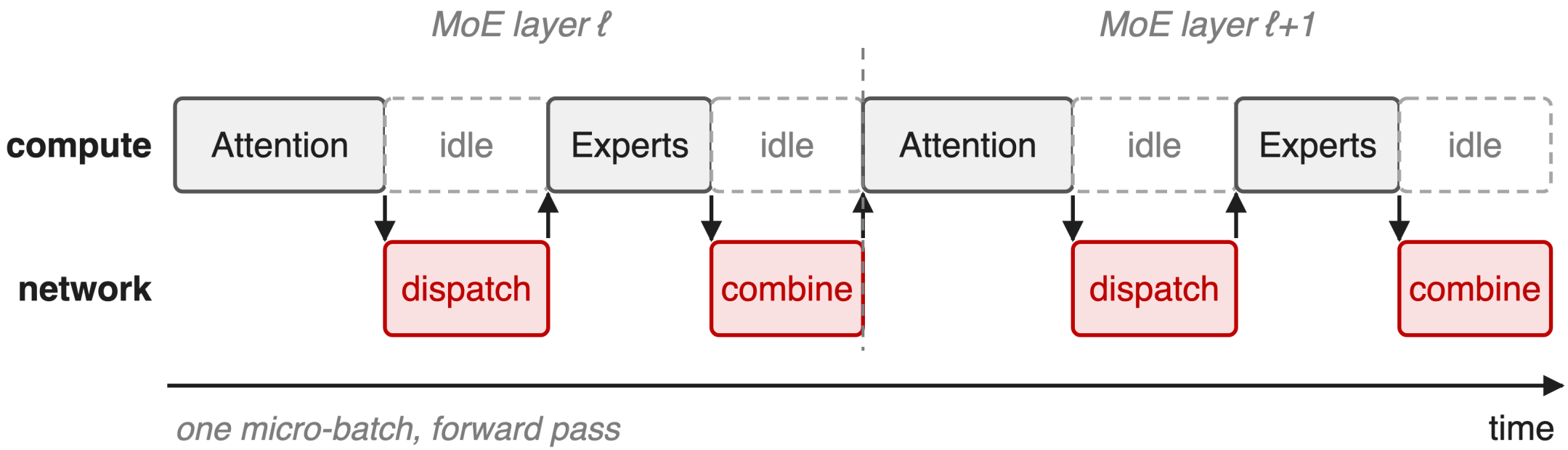
# EP all-to-all still eats a third of each step

**30–40%**

of DeepSeek-V3 iteration time is EP all-to-all on H100, even with DeepEP (40–60% before)

*Megatron-Core MoE technical report, arXiv 2603.07685, §4.2 (p.29) and §4.2.3 (p.32-33)*

# One micro-batch cannot hide its own all-to-all




**Every all-to-all sits between two computations that need its result**

# Earlier fixes shrink or hide it at a price

- **Faster dispatcher** (DeepEP)
- Fewer bytes, lower latency
- **Still 30–40%**, still on the critical path

- **Overlap schedules**
- DualPipe: bidirectional, **2× parameters**
- FWD-FWD merge: **2× activations**

**Needed: independent work at no memory cost, inside standard 1F1B**

# Insight: adjacent 1F1B passes are independent

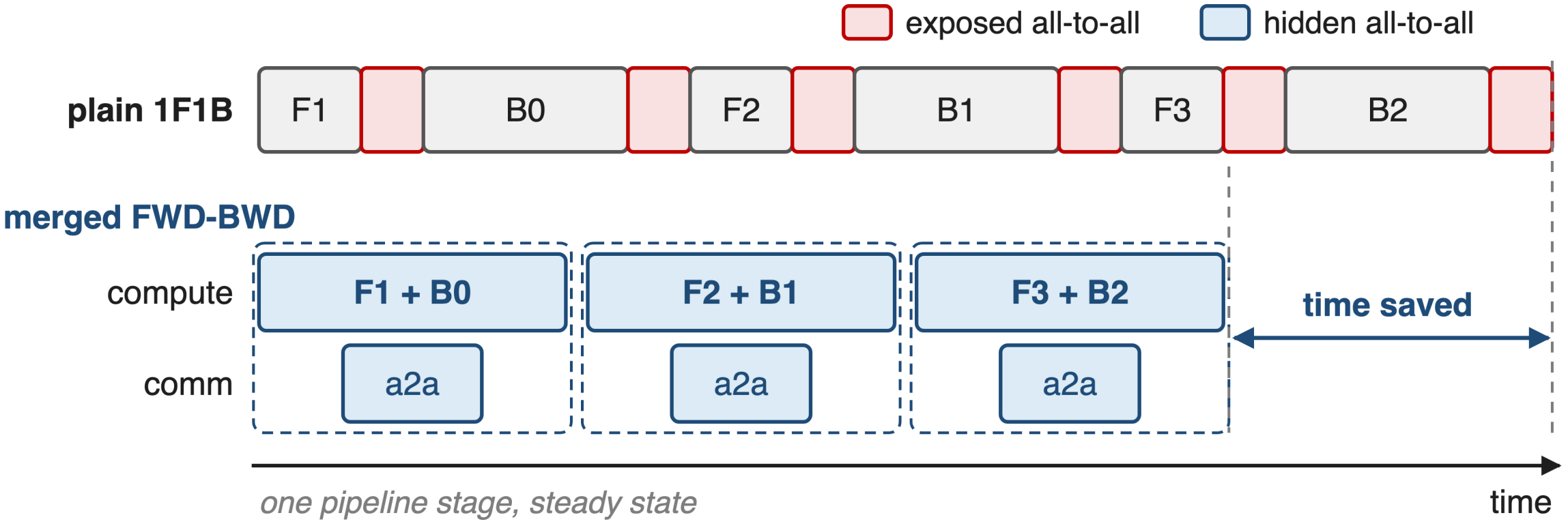
- Steady state alternates **forward of n+1** and **backward of n**
- They share no data; backward carries ~2× forward compute

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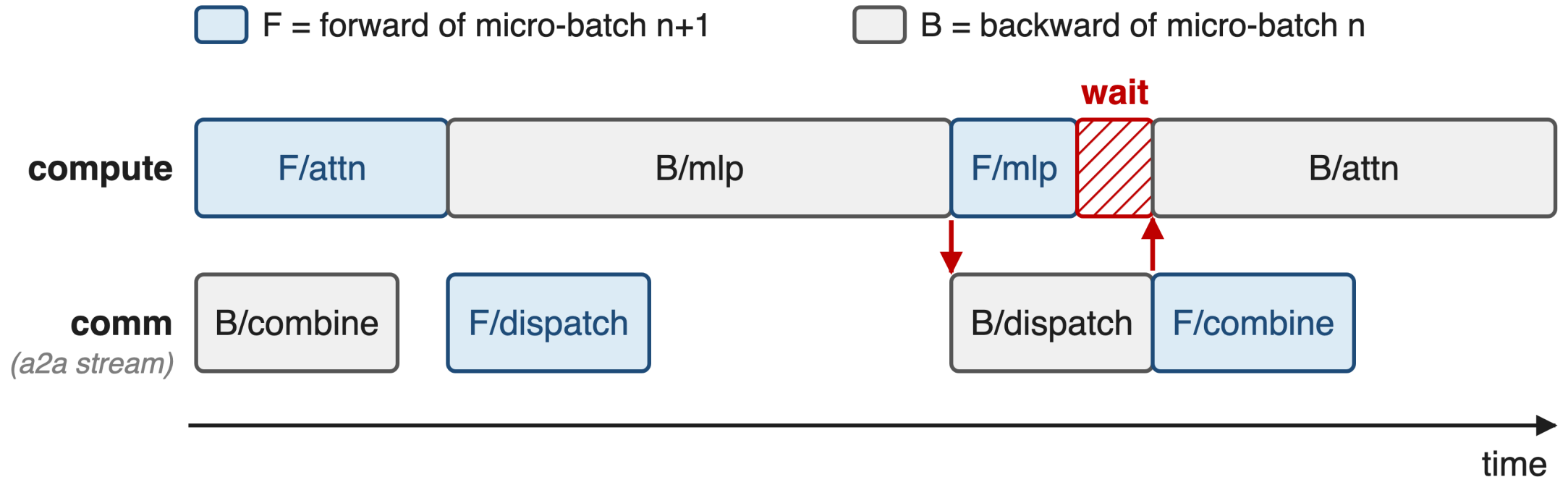
**Pair them layer by layer: each pass's all-to-all hides under the other's compute**

# Merge each 1F1B step into one F+B pair



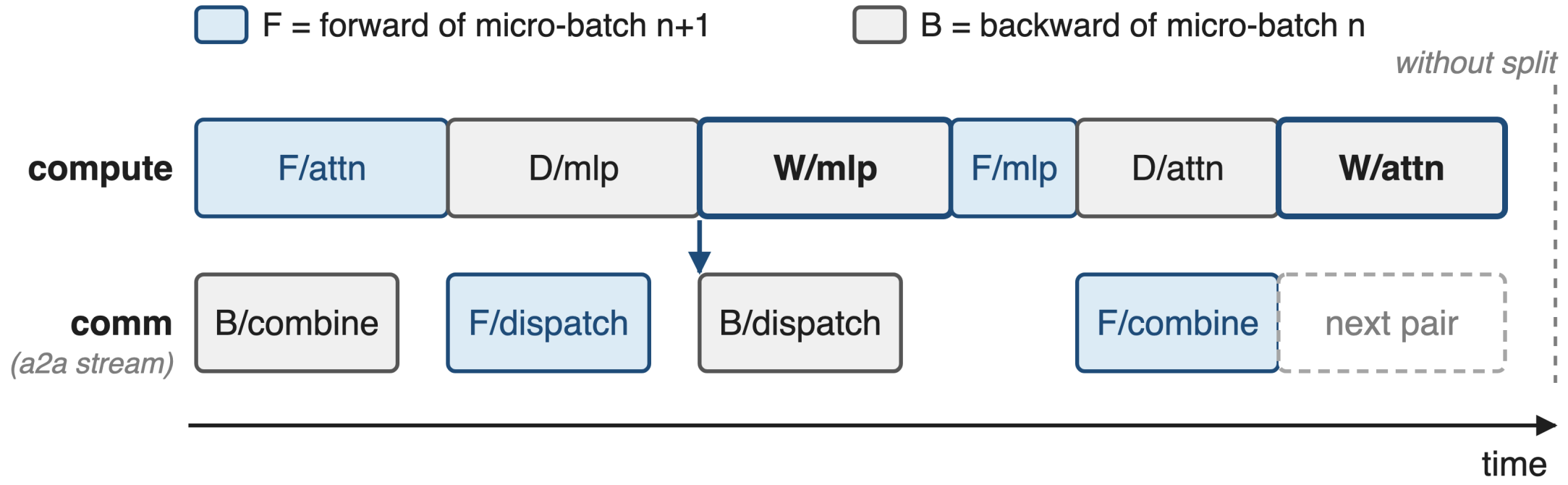

*Schematic, not to scale. Flags: --overlap-moe-expert-parallel-comm --delay-wgrad-compute (MCore v0.14)*

# ❶ Two CUDA streams swap compute and all-to-all



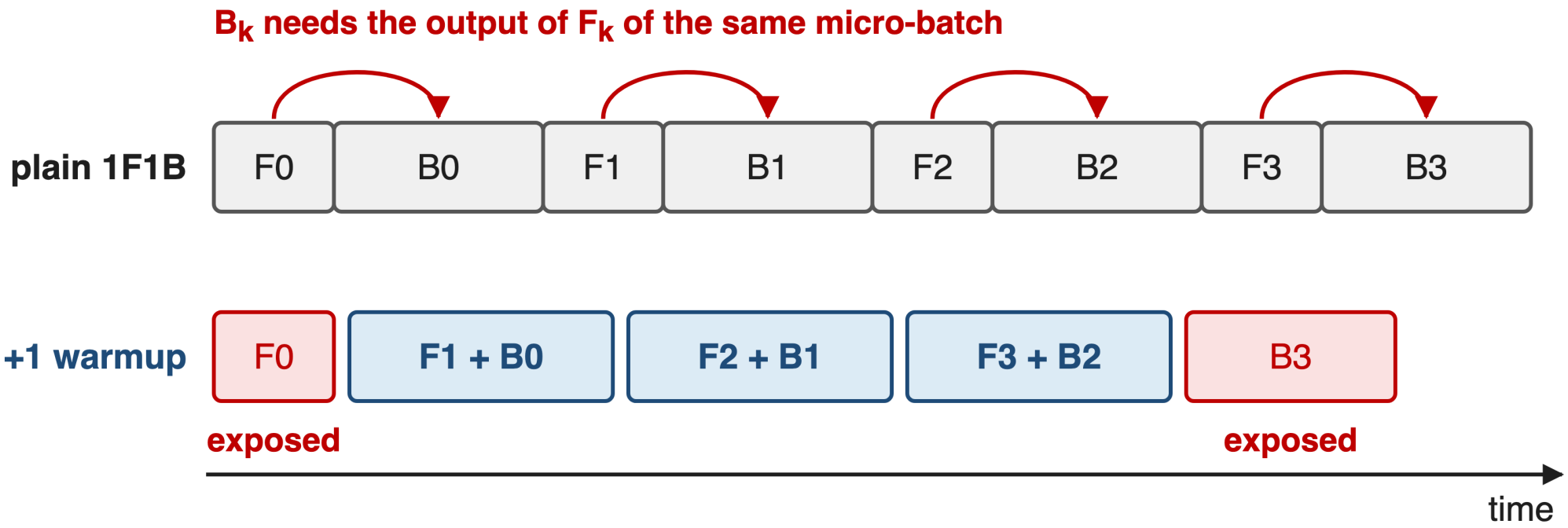

**Forward layer i of micro-batch n+1 pairs with backward layer L–1–i of n**

# ❷ W/D split releases backward dispatch early




**B/dispatch needs only D/mlp; W/mlp and the delayed W/attn fill the gaps**

# ❸ One extra warmup forward decouples every pair

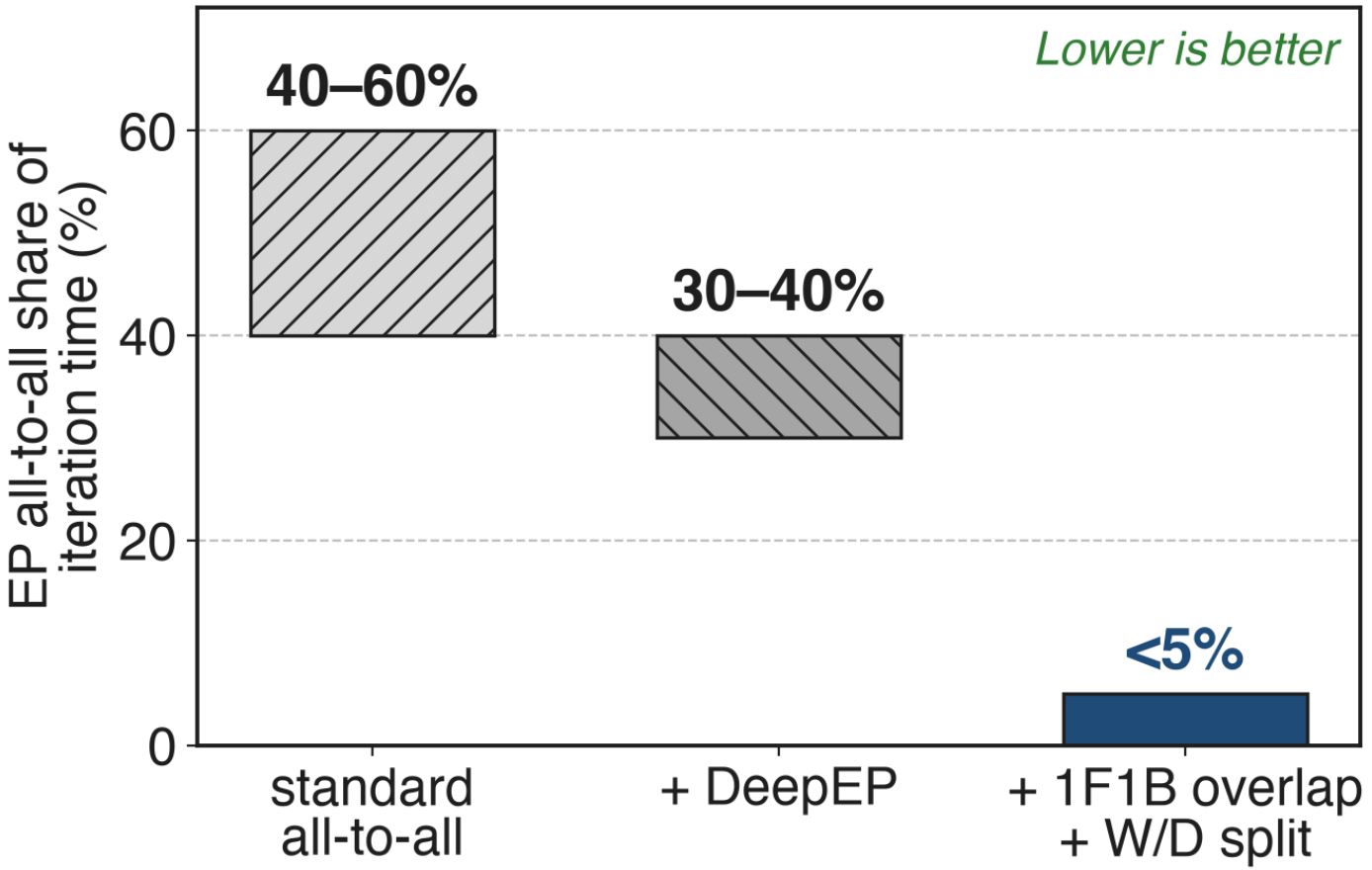



*Interleaved 1F1B adds one warmup micro-batch when overlap is on; VPP is required whenever PP > 1*

**Only the first forward and the last backward keep their all-to-all exposed**

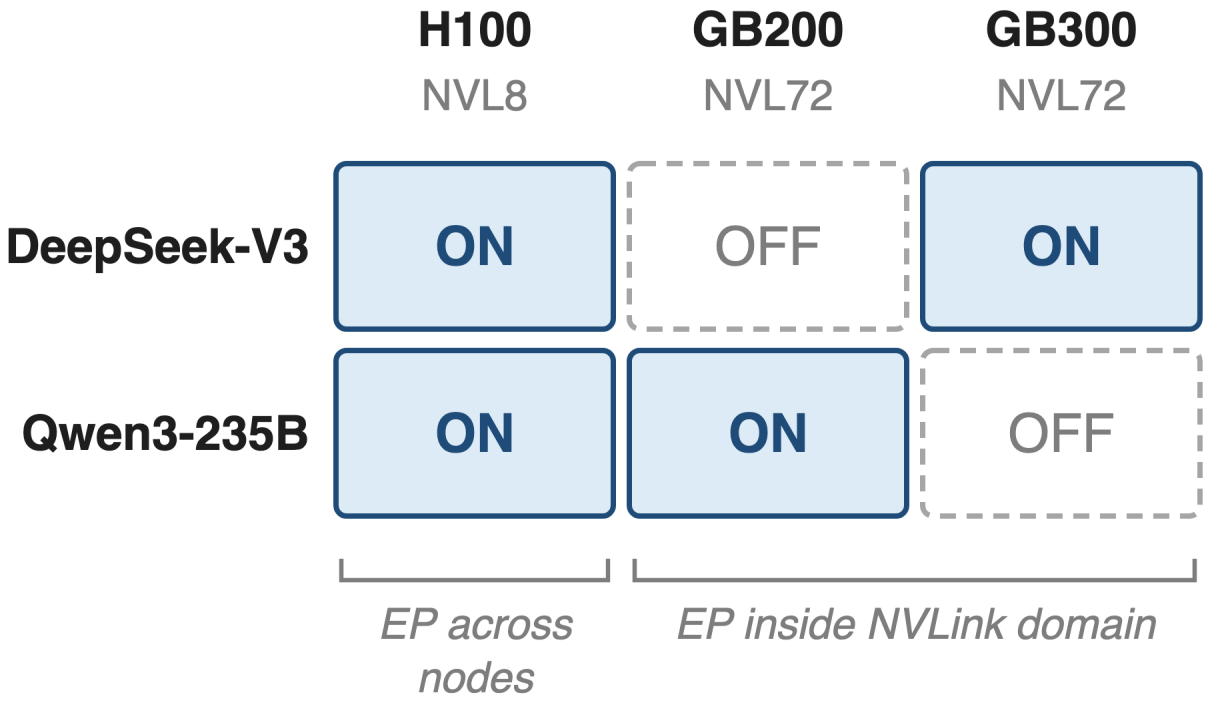
# EP all-to-all drops from 30–40% to under 5%

- DeepSeek-V3, EP64 across 8 H100 nodes
- **93%** of expert-communication latency hidden
- Whole comm-wall stack: all-to-all **< 10%**



# On H100 always on; on NVL72 a tuning choice

- H100: cross-node EP; on in the **368** TFLOPS/GPU DeepSeek-V3 run
- NVL72: EP64 stays in NVLink
- GB200 DeepSeek-V3 spends that memory elsewhere



# The cost: buffers, SMs, and exposed edges

- **Extra buffers**: on H100 only after FP8 freed memory
- **SM carve-out**: DeepEP's **20** SMs cost ~**20%** GEMM efficiency

- Needs EP > 1, VPP when PP > 1
- No full or `moe` recompute; MTP ≤ 1 layer
- Both flags off by default

# Summary

- EP all-to-all: **30–40%** of DeepSeek-V3 iteration time on H100, even with DeepEP
- **Insight**: 1F1B already runs two independent micro-batches, so pair them
- Two streams, W/D split, one extra warmup micro-batch
- **< 5%** exposed all-to-all (**93%** hidden)

**https://arxiv.org/abs/2603.07685**